

Incoming CAM / DENM Message

Communication Trust

COMMUNICATION TRUST

PKI — Identity & Certificate Validation

B1 (SCSV) — Protocol & Structural Validation

MBD — Misbehavior Detection

B2 (CSIA) — Behavioral Trust

Existing architecture (PKI, MBD, CP)

CP — Cooperative Perception

← SEMANTIC TRUST BOUNDARY →

← STBVs bypass all layers above

Semantic Trust

SEMANTIC TRUST

B3 — Semantic Trust Gate (DeBERTa-v2)

New: semantic content verified

Decision Trust

DECISION TRUST

Trust Decision Engine

Dempster-Shafer / Yager Fusion

Decision Trust Score (pignistic)

Trust-informed decisions

Accept

Caution

Reject

Planner / Vehicle Controller